INCH-POUND

MIL-C-24643/35A
14 March 1994
SUPERSEDING
MIL-C-24643/35
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3SU
(INCLUDING VARIATION LS3SA and LS3SUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in three variations: LS3SU (unarmored), LS3SA (armored) and LS3SUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

	First	- Copper conductor, AWG no. 18 (7 by 0.0152 inch), uncoated. Maximum diameter 0.050 inch.
	Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
*	Third	- Crosslinked polyethylene insulation. Nominal wall thickness 0.013 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black, one white and one red conductor for each triad.
	Fourth	- Three conductors (one black, one white and one red) cabled together with a nominal lay of 3.5 inches to form each triad.
	Fifth	- Binder tape over each triad, applied at manufacturer's option, applied helically with overlap.
	Sixth	- Braided shield of tin coated copper. Minimum coverage of 85 percent.
*	Seventh	- Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

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Construction (non-watertight) (Continued)

Eighth - The required number of triads (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

Ninth - Binder tape, applied helically with overlap.

Construction variant for type LS3SU

* Tenth - Cross-linked polyolefin jacket. (See table I for thickness).

Construction variant for type LS3SA

* Tenth - Cross-linked polyolefin jacket. (See table I for thickness).

* Eleventh - Braided metal armor.

Construction variant for type LS3SUS

Tenth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

Eleventh - Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) Separator tape with 25 percent minimum overlap.

Twelfth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

Thirteenth - Binder tape applied helically with overlap.

* Fourteenth - Cross-linked polyolefin jacket. (See table I for thickness).

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TABLE I. Details.

Military 1/ part no. M24643/35	Size 2/	Number of triads	Cable jacket thickness min avg (inch)	Diameter over Shielded core (inch nom.)	Overall diameter (inches)					
					TYPE LS3SU		TYPE LS3SA		TYPE LS3SUS	
					min	max	min	max	min	max
-01UN	3	3	0.040	0.559	0.650	0.700	0.700	0.750	0.710	0.760
-02UN	7	7	.050	.761	.840	.910	.890	.960	.900	.970
-03UN	10	10	.050	.987	1.100	1.190	1.150	1.240	1.160	1.250
-04UN	14	14	.050	1.049	1.200	1.290	1.250	1.340	1.260	1.350
-05UN	19	19	.050	1.199	1.340	1.430	1.390	1.480	1.400	1.490
-06UN	24	24	.060	1.370	1.580	1.670	1.630	1.720	1.640	1.730
-07UN	30	30	.060	1.460	1.680	1.770	1.730	1.820	1.740	1.830
-08UN	37	37	.060	1.575	1.840	1.930	1.890	1.980	1.900	1.990
-09UN	44	44	.060	1.800	2.060	2.100	2.110	2.200	2.120	2.210

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LS3SU, LS3SA and LS3SUS.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum ...	7.06
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Shield to shield (Triad shields only)	500
* Shield to shield (LS3SUS overall shields only)	200
Insulator resistance - Megohms/1000 feet minimum	
Conductor to conductor	500
* Component shield to shield	100
* Shield to shield (LS3SUS overall shields only)	100
Jacket flaws (LS3SUS only)	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum ..	30
* Capacitance unbalance - percent maximum	10

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EXAMINATION AND TESTS (continued):

	Requirements
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470) ...	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - (type LS3SU and LS3SUS only) cycles, minimum	125
Armor (type LS3SA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
* Surface transfer impedance	
Milliohms per meter, maximum	700
Emp response time - db, minimum	60
* Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
* Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 19	1000
24 through 44	500

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Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

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